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Palaeomagnetism of Upper Precambrian rocks of the La Tinta Formation, Argentina

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Summary. Palaeomagnetic data from 77 hand samples collected in sequences of marine sediments of Late Precambrian age (709 ± 24 Myr) exposed at two zones (Olavarría–Sierras Bayas and Barker) in central Argentina (37.3° S 59.8° W, La Tinta Formation) are given. After thermal cleaning and chemical leaching, the samples show normal and reversed polarity and yield a palaeomagnetic pole at 300.9° E 79.3° S ($\alpha_{95} = 4.8°$). The position of this pole supports the interpretation that Gondwana was a unit in Late Precambrian times and is consistent with the Late Precambrian glaciation (Marinoan) of south-eastern Australia.

The direction and polarity of the stable RM of samples and the change of direction of the stable RM for adjacent samples along a sequence are used to correlate La Tinta rocks within a zone and between the two zones.

1 Introduction

The definition of the geodynamic processes which affected the lithosphere in Late Precambrian–Early Palaeozoic times is a key problem in Earth Sciences and has been (Upper Mantle Project) and is (Geodynamic Project) one of the main objectives of a number of multidisciplinary and international projects. Valencio (1973) indicated that palaeomagnetic data for rocks older than Devonian from the Gondwanic continents (particularly from South America) are not enough to define, without ambiguity, the geodynamic processes which affected the Gondwana Supercontinent in early geological times. The Palaeomagnetic Laboratory of the University of Buenos Aires (Argentina) has programmed the study of sequences of South American rocks assigned to the Late Precambrian and Early Palaeozoic. Results of palaeomagnetic studies carried out with Upper Ordovician (Vilas & Valencio 1978a) and Lower Ordovician (Valencio, Vilas & Mendía 1980) rocks from Argentina have recently been presented. In this paper we present the palaeomagnetic data from two sequences of sedimentary rocks (La Tinta Formation), exposed in central Argentina and assigned to the

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Late Precambrian. We also compare the position of the palaeomagnetic pole for La Tinta rocks with the positions of the palaeomagnetic poles of Precambrian age from the other Gondwanic continents in order to define the distribution of these continents in that geological time.

The correlation of Precambrian rocks is frequently complicated, because they normally have no characteristic fossil assemblages. The direction and polarity of the remanent magnetism may be used, under certain conditions, as an element for the correlation of Precambrian rocks. In this paper we use the palaeomagnetic data to establish a tentative correlation between some of the La Tinta rocks.

2 Geological evidence and sampling

Flat lying marine sediments included in the La Tinta Group (Amos, Quartino & Zardini 1972) are exposed in the central part of the Province of Buenos Aires, about 300 km south of Buenos Aires. These rocks rest on a basement, in which mineral ages have been determined to be in the range 10^3–2×10^3 Myr, and are cut by basic intrusives dated as Ordovician (Rapela, Dalla Salda & Cingolani 1974). An age which varies from Late Precambrian to Early Palaeozoic has been assigned to different exposures of the La Tinta rocks. In particular Amos, Quartino & Zardini (1972) have suggested a Precambrian age for exposures close to Olavarría (La Tinta Formation) and a probable Precambrian or Early Palaeozoic age for exposures south-east of Tandil (Balcarce Quartzites), and Zalba (1977) suggested a Precambrian age for exposures close to Barker (La Tinta Formation).

Quartzites, dolomites, pelites and limestones included in La Tinta Formation are exposed close to Olavarría – Sierras Bayas (37°S 60.2°W) (Marchese & Di Paola 1975). These rocks are normally covered by quarternary soil; however, the complete succession of La Tinta rocks in this area was integrated on the basis of geological studies carried out on different exposures. These exposures are frequently quarries where rocks are extracted for industrial purposes. The succession of La Tinta rocks in Olavarría – Sierras Bayas is integrated by the following lithosomes, that is in chronological order from the base to the top: quartzites, dolomites, quartzites, pelites and limestones (Marchese & Di Paola 1975). These authors have suggested a Late Precambrian (Late Rifean to Vendian) age for these rocks, based on the lithologic association and the presence of stromatolites. They have also indicated that the characteristics of La Tinta sediments are similar to those of Precambrian sediments from Brazil, South Africa and Australia.

Rubidium – strontium age determinations were carried out on La Tinta rocks from different quarries close to Olavarría (Bonhomme & Cingolani 1978). Samples collected from Cerro Negro (site E) and L.O.S.A. (site C) quarries yield an isotopic Rb–Sr line with an age of 709 ± 24 Myr. Samples from the AUST quarry yield ages of 797 ± 18 Myr and 756 ± 22 Myr. These ages are consistent with the stratigraphic relationship between the sediments from those quarries (Bonhomme & Cingolani 1978) and the Precambrian age suggested for the La Tinta rocks close to Olavarría by Amos, Quartino & Zardini (1972) and Marchese & Di Paola (1975).

Quartzites, sandstones, dolomites, pelites and limestones of Late Precambrian age, included in the La Tinta Formation, are exposed close to Barker (37.6°S 59.3°W, about 115 km from Olavarría; Zalba 1977). Chronologically, quartzites, brown sandstones and quartzites, are exposed at Cerro El Sombrerito from the base to the top; the lower and upper quartzites have been correlated with the lower and upper quartzites, respectively, from Olavarría – Sierras Bayas (Zalba 1977). Dolomites and quartzites, again in chronological order from the base to the top, are exposed at Cerrito La Cruz; the dolomites were correlated

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with the brown sandstones from Cerro El Sombrerito and the quartzites with the upper quartzites of that site, (Zalba 1977).

The rocks of the La Tinta Formation exposed close to Olavarría–Sierras Bayas were sampled at five different sites: Tofoletti quarry (A); the road between L.O.S.A. and La Alcancía quarries (B); L.O.S.A. quarry (C); La Alcancía quarry (D) and Cerro Negro quarry (E). Forty-seven hand samples were collected at different stratigraphic levels at these sites.

The Late Precambrian rocks of the La Tinta Formation exposed close to Barker were sampled at four different sites: Cerro El Sombrerito (F, lower and upper quartzites); Cerrito La Cruz (G, dolomites and quartzites); El Infierno quarry (H, limestones) and Caravelli quarry (I, pelites). Thirty hand samples were collected at different stratigraphic levels at these sites.

Hand samples were oriented using both solar and magnetic compasses. Observations of bedding plane were carried out in each of the nine sampling sites.

3 The palaeomagnetic study

The samples from quartzite exposures are extremely hard; therefore, in most cases, only one specimen was cut from each of these samples. One hundred and thirty-eight cylinders of 2.5 cm diameter and height were cut. A spinner magnetometer was used to measure the remanent magnetization. The intensity of natural remanent magnetization (NRM) of two hand samples from Olavarría – Sierras Bayas was extremely weak; these samples were not included in the palaeomagnetic study. Detailed thermal demagnetization was used to isolate the stable remanence of one pilot cylinder of each sample (in steps of 100°C up to 600°C). A small secondary component of remanence was present in the majority of samples; it had the characteristics of a viscous remanent magnetization and was destroyed by the thermal demagnetization. Fig. 1 shows the changes in direction and intensity of the magnetic remanence of pilot specimens from La Tinta rocks in response to thermal treatment. The optimum temperature varied between 200 and 400°C, at which more than 50 per cent of the NRM was retained in the quartzites (Fig. 1a) and dolomites (Fig. 1b), and 20 per cent in limestones (Fig. 1c) and pelites (Fig. 1d). One other specimen from the samples from dolomites, limestones and pelites was submitted to the optimum temperature in order to improve the definition of the direction of the stable remanence of each sample. We were unable to isolate any stable remanence in one of the hand samples.

Quartzites were also submitted to chemical leaching in those cases where two or more specimens were cut from hand samples. The specimens were immersed in hydrochloric acid for a given time (17, 40, 80, 175, 350, 600, 1000 and 1500 hr), washed in water, measured and then re-immersed in acid (Roy & Park 1974). The optimum demagnetizing chemical leaching time varied between 600 and 1000 hr at which about 50 per cent of the natural intensity was retained (Fig. 1a). In the majority of samples no substantial differences between the directions of the stable remanence isolated during the thermal and chemical cleaning were observed (Fig. 1a). In a few samples chemical leaching was more effective than thermal treatment.

Particular mention should be made of the results obtained from sample number 40 (dolomite). Two specimens were cut from this sample; they were named 40a and 40b. Specimen 40a came from a higher stratigraphic position in the sample than 40b. Specimen 40a gave northerly declination and upward dipping inclination ($D = 24.5°$, $I = -64.5°$); whereas 40b gave southerly declination and downward dipping inclination ($D = 207°$, $I = 65°$). The direction of the stable remanence of specimen 40a was consistent with that of sample 41 ($D = 1°$, $I = -64°$), collected from a stratigraphic position 2.5 m higher than specimen 40 (Fig. 3). The direction of the cleaned remanence of specimen 40b was consistent

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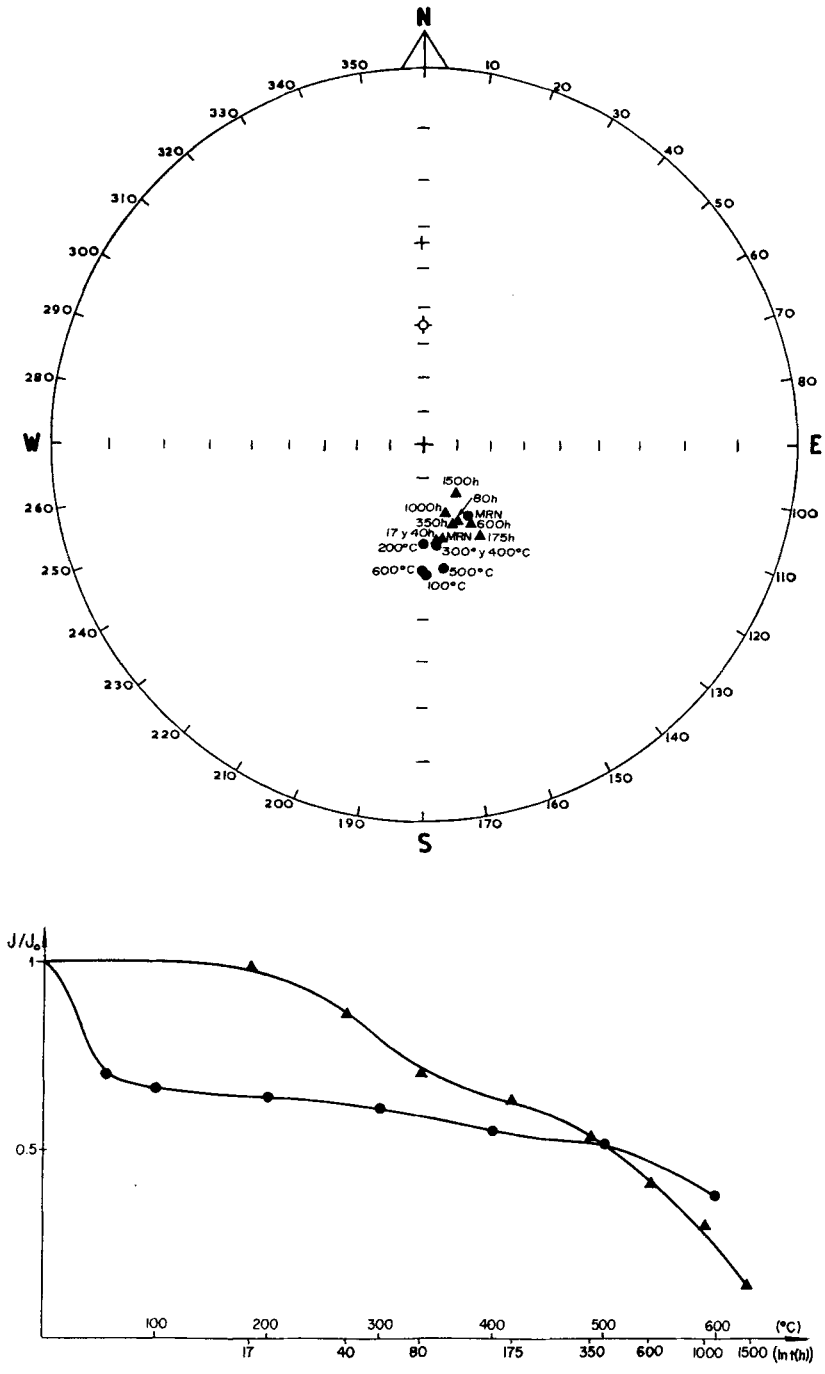


Figure 1 (a)

Figure 1. Changes in direction and intensity of NRM of the pilot specimen (○) and one other cylinder (□) from samples from (a) La Tinta quartzites, (b) dolomites, (c) limestones and (d) pelites in response to thermal demagnetization. In (a) the changes of NRM (△) are also shown for one cylinder from quartzites in response to chemical leaching. Solid symbols indicate downward dipping directions. The directions of present geomagnetic field and of the axial dipole are shown by + and ⊹ respectively.

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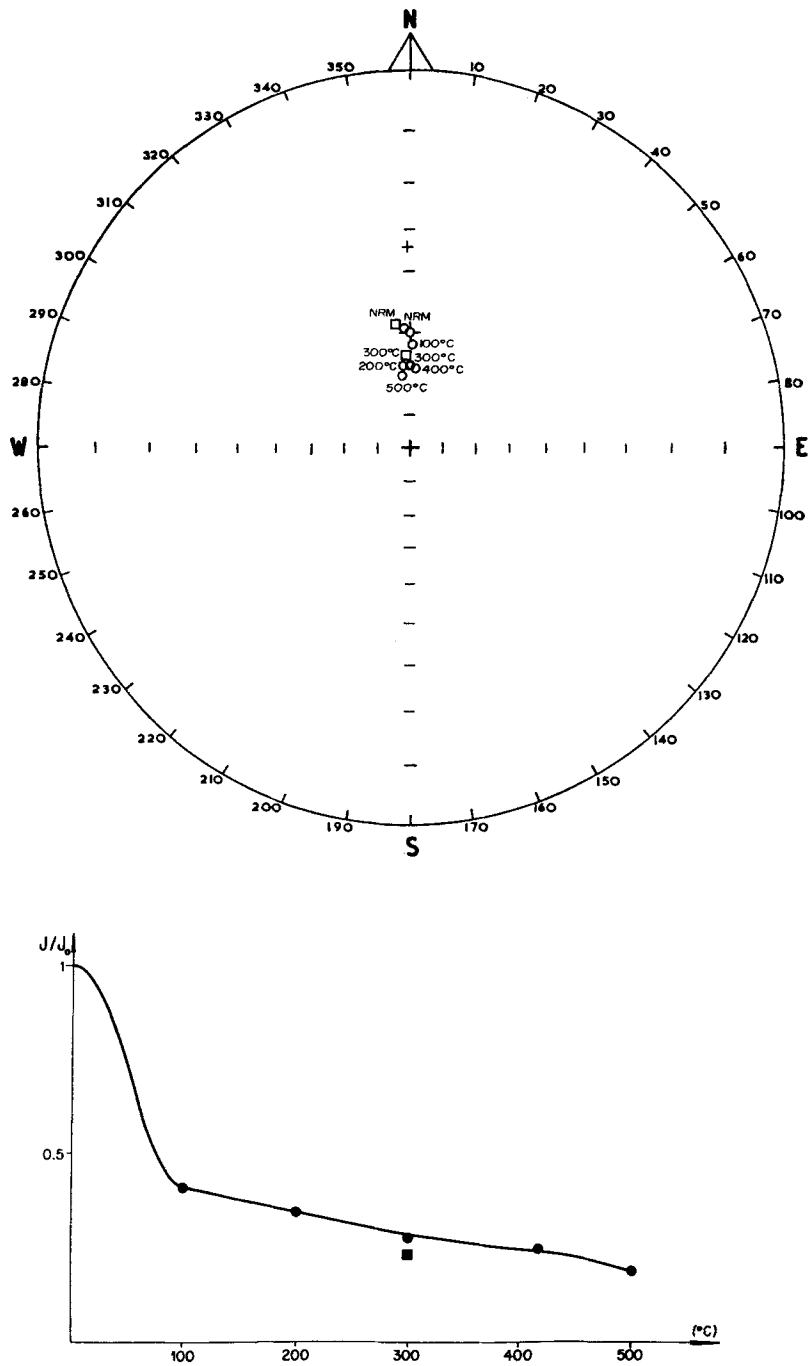


Figure 1(b)

with the direction of the cleaned remanence of sample 39 ($D = 191°, I = 57°$) collected from a stratigraphic position 2.5 m lower than specimen 40 (Fig. 3). This shows that two directions of stable remanence were isolated in sample 40; these directions record a reversal of the geomagnetic field. Specimens from samples 7, 20 and 22 also yielded different directions of stable remanence.

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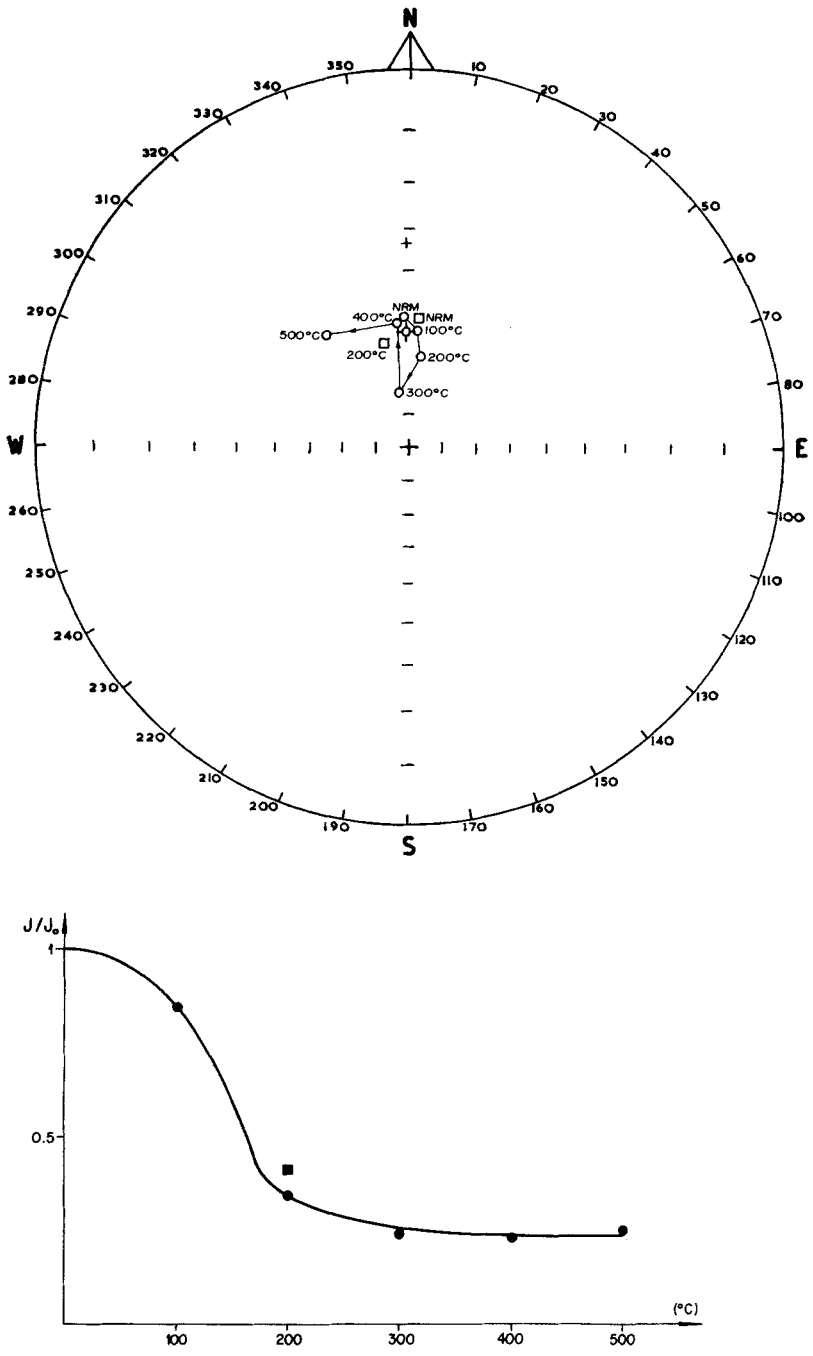


Figure 1(c)

The method of Fisher (1953) was used to calculate mean directions of NRM and stable remanent magnetization of each hand sample by assigning unit weight to the direction of the natural and stable remanence of each of its specimens, respectively. Hand sample directions of remanence were referred to the palaeohorizontal; however the fold test could not be applied because all the samples were collected from sites where the dip of strata is very

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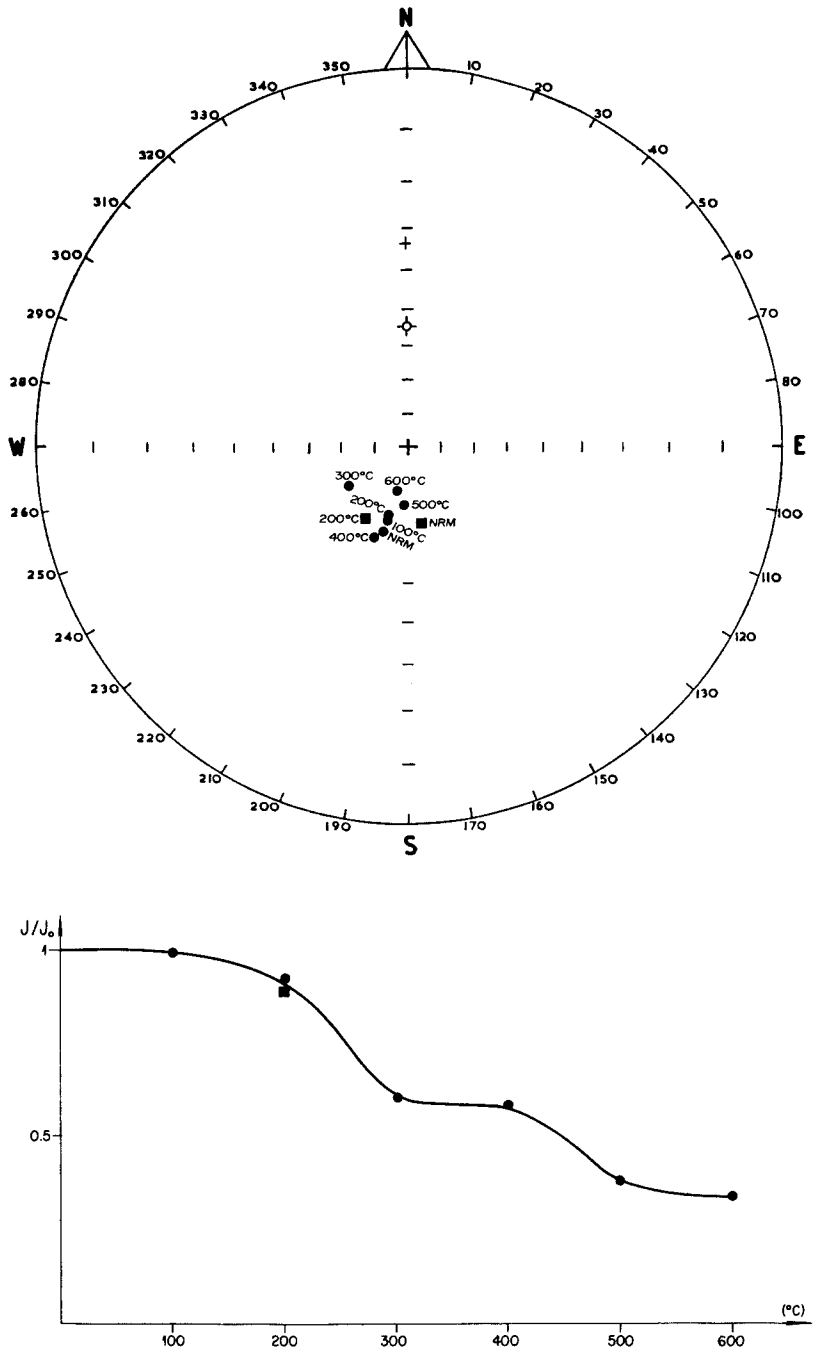


Figure 1(d)

small. Table 1 shows the bedding plane data for the different sites at Olavarría–Sierras Bayas.

The mean directions, after bedding plane correction, of cleaned RM of each hand sample from the La Tinta Formation from Olavarría–Sierras Bayas and Barker are shown in Fig. 2(a) and (b), respectively.

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Table 1. Site latitudes and longitudes and bedding-plane data for La Tinta rocks from Olavarría–Sierras Bayas.

Site (quarry)	Lat (°S)	Long (°W)	N	Strike (°)	Dip (°)	α_{95} (°)	k
Tofoletti (A)	36.90	60.25	4	151	8	7.6	102
L.O.S.A.– La Alcancía (B)	36.95	60.25	5	96	7	4.6	274
L.O.S.A. (C)	37.00	60.30	6	59	2	7.8	74
La Alcancía (D)	36.90	60.20	7	66	4	1.7	1143
Cerro Negro (E)	37.00	60.20	12	130	13	2.8	245

Virtual geomagnetic poles (VGP) for each hand sample of the La Tinta Formation were computed. A mean pole was computed by taking the mean of these VGP's; VGP's situated more than 40° from the mean pole were rejected and a new mean pole was computed. A group of VGP's, all within 40° of the mean pole, was obtained. This group yielded a palaeomagnetic pole for the La Tinta Formation at 300.9° E 79.3° S (SAPrϵ_2, $N=53$, $\alpha_{95}=4.8°$, $k=16.9$). The pole for the La Tinta rocks from Olavarría–Sierras Bayas was 303.7° E 81.6° S ($N=37$, $\alpha_{95}=5.1°$, $k=22$) and from Barker, 295.1° E 69.5° S ($N=16$, $\alpha_{95}=9.9°$, $k=14.7$). The rejected VGP's were classified as oblique.

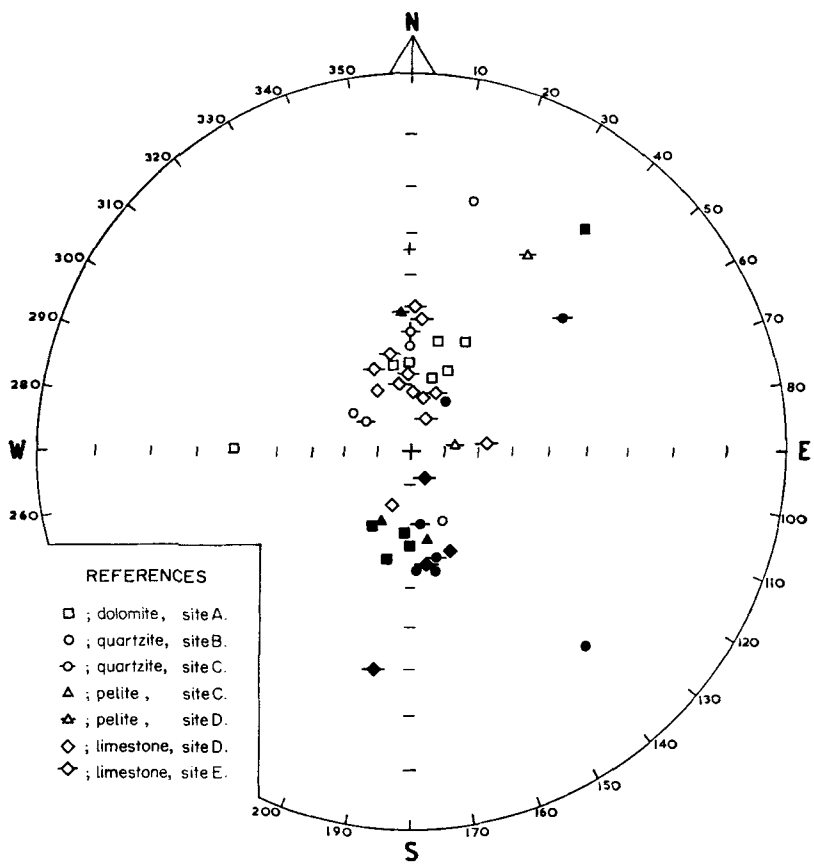


Figure 2(a)

Figure 2. The La Tinta Formation. Directions of cleaned RM, referred to the palaeohorizontal, of hand samples from (a) Olavarría–Sierras Bayas and (b) Barker. The key to the symbols is as given in Fig. 1.

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3 Interpretation of results

The normal and reversed stable RM isolated in samples from the La Tinta rocks from Olavarría–Sierras Bayas and Barker, as well as the two directions of stable RM isolated in some of these samples, suggest that this remanence was acquired during or shortly after the deposition of those rocks.

The composite magnetic stratigraphy for the La Tinta Formation exposures close to Olavarría–Sierras Bayas, consisting of logs of sample-mean values of stable declination and inclination, is plotted alongside the lithologic log in Fig. 3. The log of the angle between the VGP for each hand sample and the mean polar position is also shown in this figure. Logs for the different sites are plotted in stratigraphical order. However, the geological data do not show if there is overlapping of, or a gap between, the successions. We interpret a succession of oblique VGP's as indicating a reversal of the polarity of the geomagnetic field within a sequence of rocks of a given polarity if at least one VGP lies within 40° of the palaeomagnetic pole of opposite polarity (Valencio, Vilas & Mendía 1977).

Fig. 3 also shows the magnetic stratigraphy for La Tinta rocks exposed at El Infierno (site H) and Caravelli (site I) quarries, close to Barker.

The correlation of Precambrian rocks, even within a basin, is one of the main problems in earth sciences. All data should be used to establish this correlation. Therefore, the comparison of the direction and polarity of the stable RM may be used to help correlation of

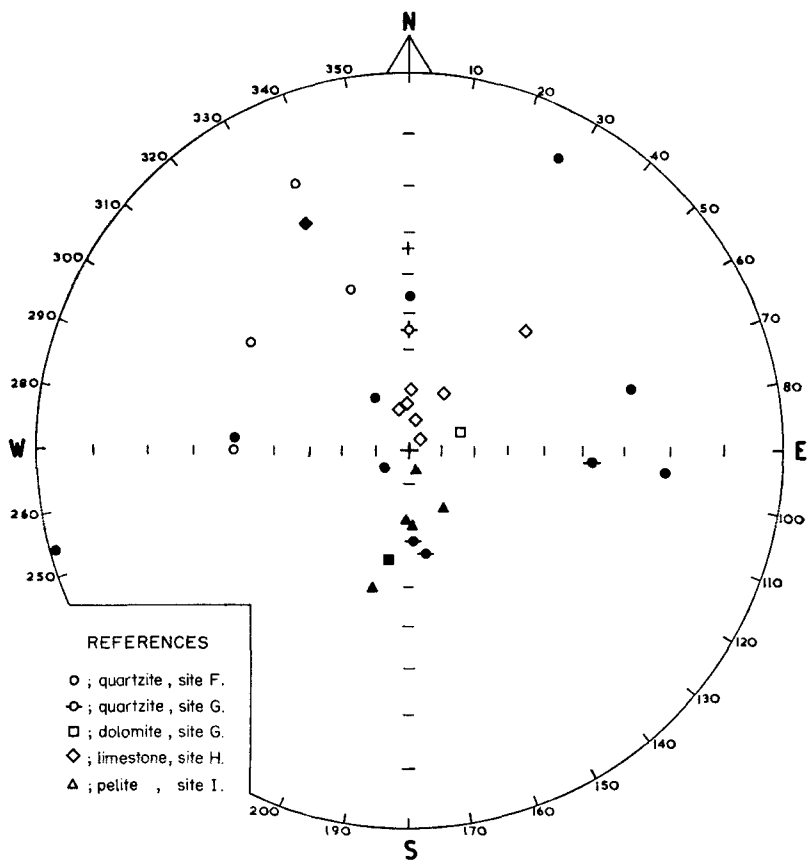


Figure 2 (b)

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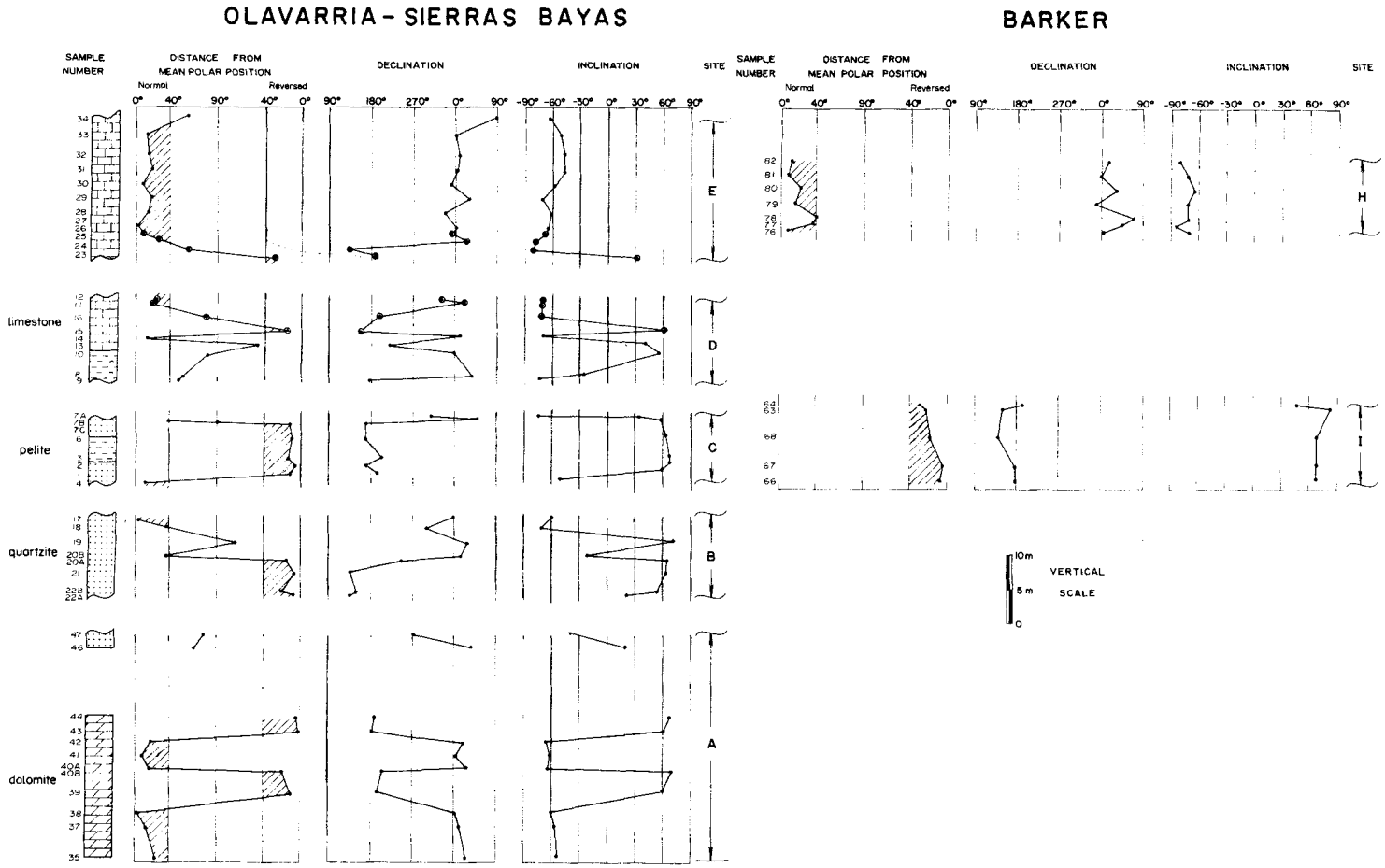


Figure 3. Mean declination and inclination of stable remanence, the angle between the VGP for each stratigraphic level and the palaeomagnetic pole through the sequence of La Tinta rocks exposed close to Olavarría–Sierras Bayas and Barker, Province of Buenos Aires.

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Precambrian rocks. This application is more effective when samples for the palaeomagnetic study are collected in stratigraphic order. In this case the correlation should also be based on comparison of the change in the direction of the stable remanence for adjacent samples along the sequence. For this application in our study we must compare palaeomagnetic data from a given lithosome. An example is the case recorded in the highest samples (15, 16, 11, 12) from site D (La Alcancía quarry, limestones) and the lowest samples (23, 24, 25, 26) from site E (Cerro Negro quarry, limestones) in Fig. 3. The palaeomagnetic data strongly suggest a correlation between the sections of sequences D and E (Olavarría–Sierras Bayas) from where these samples were collected.

The comparison of directions of cleaned RM of hand samples from Olavarría–Sierras Bayas (a) and Barker (b) (Fig. 2) suggests that they acquired their RM under the same

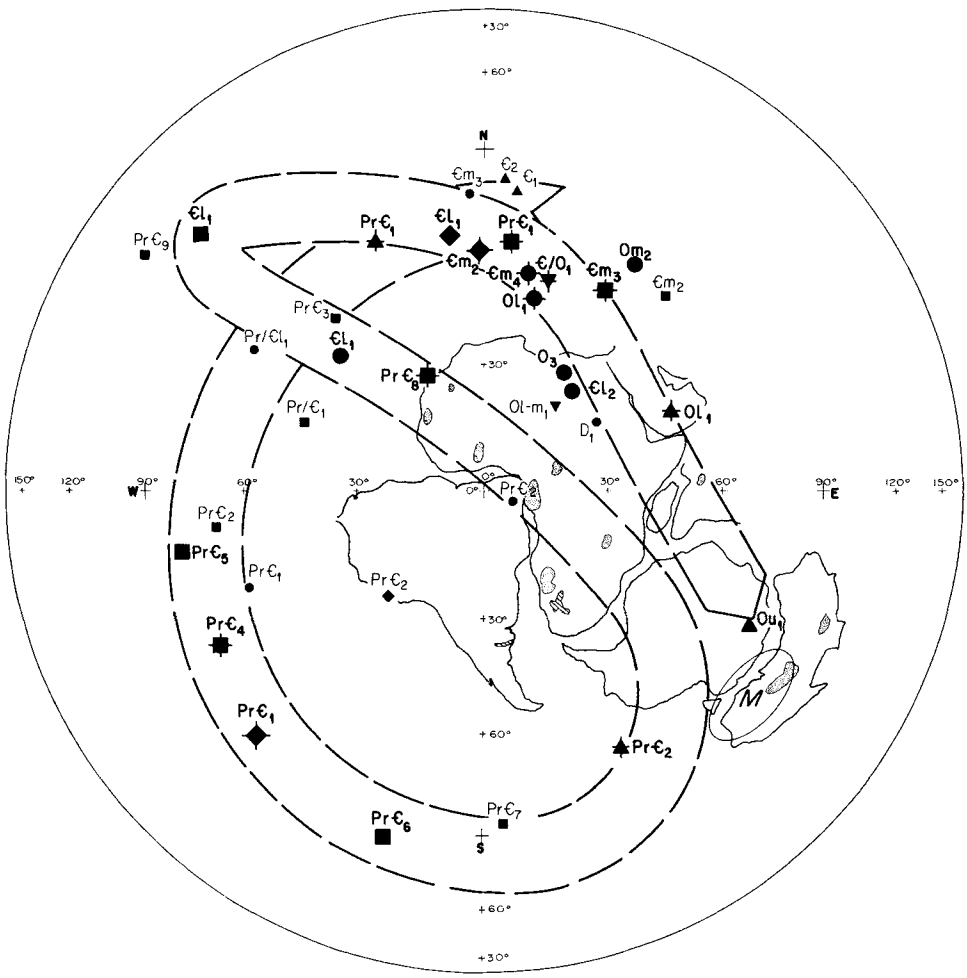


Figure 4. Pole position for the La Tinta Formation (SAPrϵ_2, 709 ± 24 Myr) plotted on the Vilas & Valencio (1978b) reconstruction. Pole positions are those quoted in Vilas & Valencio (1978a). ▲, South America; ■, Africa; ●, Australia; ◆, India; ▼, Antarctica. Key: ⯅, six points (key pole); ▲, five points (main pole); ▲, four or three points (subsidiary poles). Lambert's equal-area projection. *M* is the locus of possible pole position for the Marinoan glacial beds (McElhinny & Embleton 1976). ▨, Precambrian marine sediments for La Tinta (South America) and Nama Groups (Africa). ⬚, Late Precambrian glaciations (after McElhinny & Embleton 1976).

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geomagnetic field. That is, the palaeomagnetic data suggest that these rocks have the same magnetic age and establish a correlation between them. In particular, the magnetic stratigraphy for the pelites from site I (Caravelli quarry, Barker) and site C (L.O.S.A. quarry, Olavarría–Sierras Bayas), and for the limestones from site H (El Infierno quarry, Barker) and site E (Cerro Negro quarry, Olavarría–Sierras Bayas), respectively, suggest a correlation between them (Fig. 3).

Following the classification system suggested by Vilas & Valencio (1978a) we have allocated six points (key pole) to the palaeomagnetic pole for the La Tinta Formation ($SAPr\epsilon_2$, 709 ± 24 Myr). Fig. 4 shows the position of $SAPr\epsilon_2$ plotted on the Vilas & Valencio (1978b) reconstruction of Gondwana. The figure also shows the palaeomagnetic pole positions for the Gondwanic continents and the joint apparent polar wandering path (swathe) for Gondwana for the period 1000–400 Myr suggested by Vilas & Valencio (1978a).

The examination of Fig. 4 shows that the position of $SAPr\epsilon_2$ is consistent with the apparent polar wandering curves for Gondwana suggested by McElhinny & Embleton (1976) and Vilas & Valencio (1978a) and contributes to a better definition of the common polar swathe for this supercontinent. In particular, the figure shows that the position of $SAPr\epsilon_2$ is consistent with the position of palaeomagnetic poles for the Mbozi Complex ($AfPr\epsilon_6$, 743 ± 30 Myr, 5 points, main pole) and the Pre Nama dykes ($AfPr\epsilon_7$, 653 ± 70 Myr, 4 points, subsidiary pole) (references quoted in Vilas & Valencio 1978a). Fig. 4 also shows that the position of $SAPr\epsilon_2$ (709 ± 24 Myr) is consistent with the locus (M) of the possible pole position for the Late Precambrian Marinoan glacial beds (about 700 Myr ago, southern Australia) (McElhinny & Embleton 1976); we should note that the position of $SAPr\epsilon_2$ is in between the position of $AfPr\epsilon_6$ (743 ± 30 Myr) and the zone affected by the Marinoan glaciation. Therefore, the position of $SAPr\epsilon_2$ supports the interpretation that Gondwana was a unit at least as far back as 1000 Myr ago (Vilas & Valencio 1978a). Particularly, this means that South America and Africa were joined by their littorals in Late Precambrian time. This is consistent with the similar characteristics of some Upper Precambrian sediments from the southern part of those continents. This was observed by Du Toit (1927) who suggested that the non-fossiliferous sequences, composed of quartzites, shales and limestones, from South America and Africa belong to the same Late Precambrian system. More recently, the La Tinta Group rocks were correlated with rocks from the lower part of the Nama Group (Late Precambrian to Cambrian, southern Africa, Fig. 4) by Dalla Salda (1979), who suggested that the sediments included in these groups were accumulated in a common, shallow, marine basin.

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